

Deterministic Augmentation Beats LLM Distillation: Thai G2P from 2.32% PER with Rule-Based Labels

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Abstract

Thai grapheme-to-phoneme conversion is bottlenecked by labeled data: high-quality dictionaries cover $\sim 10\text{K}$ entries. The prevailing 2026 recipe for low-resource tasks is distillation from large language models. We show this is unnecessary and unsafe for phonological tasks: LLMs hallucinate tones and phonotactics just as they hallucinate haraqat in diacritization. Instead, we augment with a *deterministic* rule-based phonemizer (epitran) over 50K unlabeled Thai Wikipedia sentences and continue fine-tuning a Thai-pretrained umt5-small. PER falls from 3.24% to **2.32%** on a fixed 1,219-sentence Kaikki-derived test set—surpassing the public baseline (B-K/umt5-thai-g2p-v2-0.5k, 6.37% on the same set) by $2.7\times$. We report extensive negative results: curriculum learning, output-vote ensembles, and from-scratch seq2seq/CTC architectures all fail, locating the bottleneck precisely in labeled-data coverage.

1 Introduction

Thai script omits vowel and tone information that phonology requires; G2P must supply it. Published Thai G2P models (B-K/umt5-thai-g2p) are trained on dictionaries of $\sim 10\text{K}$ entries.

Contributions:

1. **Deterministic augmentation:** epitran over 50K Wikipedia sentences $\rightarrow$ PER 3.24% $\rightarrow$ **2.32%**, no LLM, no cost, no hallucination.
2. **LLM unsuitability documented:** general LLMs cannot reliably produce Thai phonology (tone hallucination), mirroring the haraqat hallucination known for diacritization.
3. **Bottleneck localization** via ablations: from-scratch seq2seq collapses; CTC plateaus at $\sim 70\%$ PER; curriculum *hurts* at the data-limited optimum; output-vote ensembles are flat.

2 Related Work

B-K/umt5-thai-g2p-v2-0.5k: public Thai G2P (our base). ByT5 [1]: byte-level multilingual pre-training. 2026 LLM-distillation practice (e.g. DeepSeek-V4 “ten teachers”) motivates our negative result.

3 Method

Continue fine-tuning B-K/umt5-thai-g2p-v2-0.5k (300M) for 10 epochs on 9.7K Kaikki Thai-IPA pairs, LR 5×10^{-5} , beam 4. Then a second stage on the union with 50K epitran-labeled Thai

Wikipedia sentences (epitran **tha-Thai**; filtering: Thai-script ratio, length bounds; 143,979 raw lines skipped, 50K kept).

4 Experiments

All numbers: 1,219-sentence Kaikki-derived test, PER at word level.

Configuration	PER
B-K/umt5-thai-g2p-v2-0.5k (public baseline)	6.37%
Our continued FT (Kaikki 9.7K)	3.31%
+ seed selection (s789)	3.24%
+ epitran augmentation (60K total)	2.32%
<i>Negative results:</i>	
From-scratch ByT5-small	13%
From-scratch seq2seq	collapse
CTC	~70% plateau
4-seed medoid-vote ensemble	3.26%
Curriculum (3 phases, HEBATRON-style)	3.60%
20 epochs (vs 10)	4.05%

5 Why not LLMs?

Phonological labeling demands exactly the property LLMs lack: faithfulness to a system of rules. Thai’s five phonemic tones and complex grapheme-phoneme correspondence invite fluent-sounding but wrong output—the same failure documented for Arabic haraqat. Deterministic phonemizers (epitran, eSpeak) never hallucinate; their systematic errors are learnable biases that a fine-tuned model corrects, as our 2.32% result demonstrates.

6 Discussion

Language-specific pretraining over scale: Thai-exposed umt5-small (300M) beats from-scratch ByT5-small by 10 points; deterministic labels at scale close the remaining gap.

7 Limitations

Kaikki-format test; epitran labels are systematically biased (no tone marks) and the model must reconcile two label distributions. Sentence count of the test set is modest (1,219).

8 Reproducibility

interscript/secryst: augmentation (`augment_thai_epitran.py`), training (`train_thai_combined.py`), ablation scripts, and docs/RESULTS.md with all run IDs.

References

- [1] Xue et al. ByT5. 2022.
- [2] Mortensen et al. Epitran. 2016.